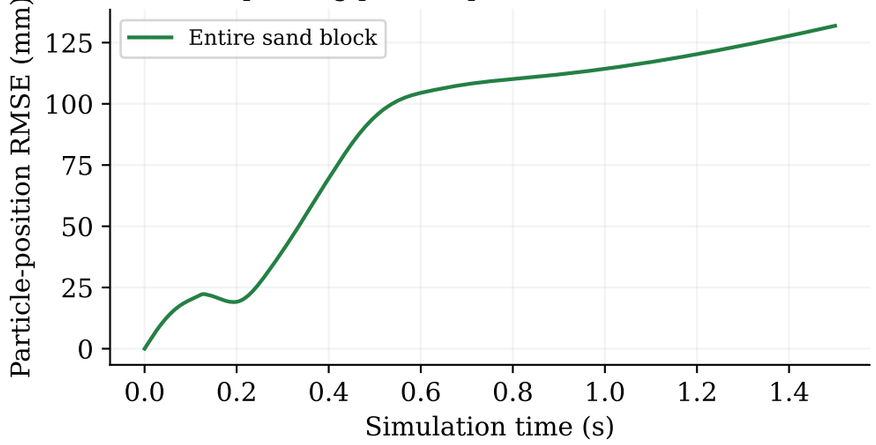
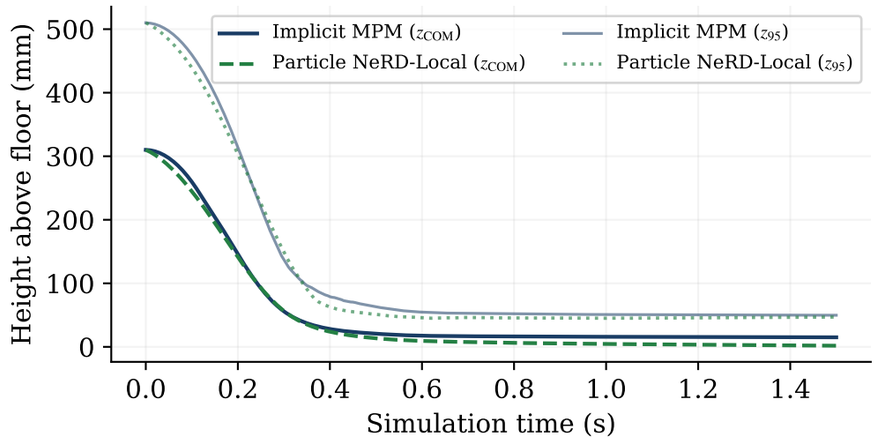
Free-Fall Sand & Granular Column Collapse Evaluation: Implicit MPM vs. Shared Particle NeRD-Local

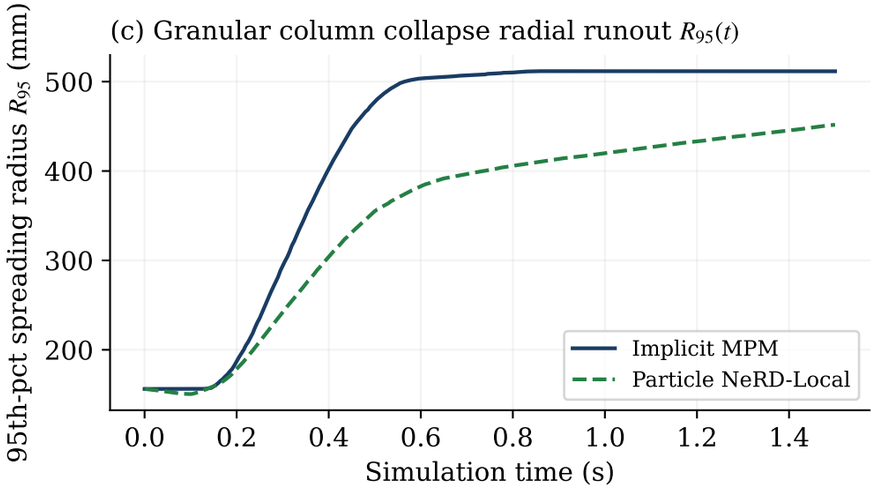
(a) Corresponding-particle position RMSE



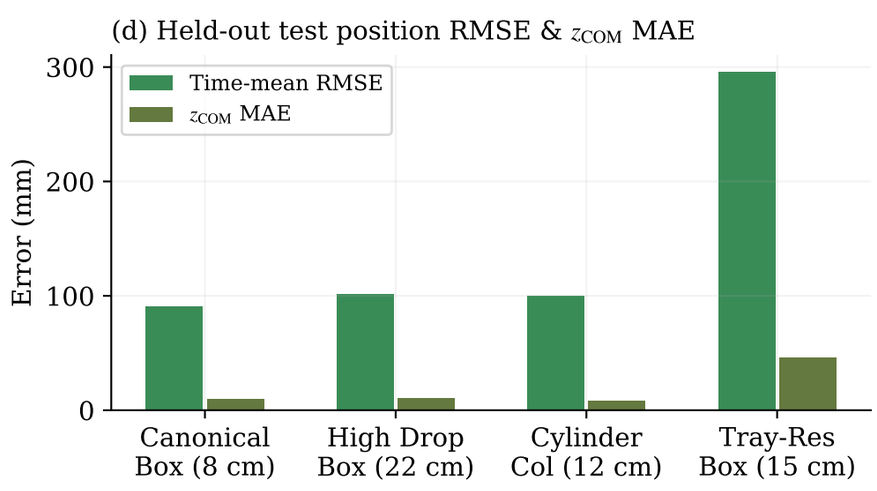
(b) Free-fall & slump vertical profile (z_{COM}, z_{95})



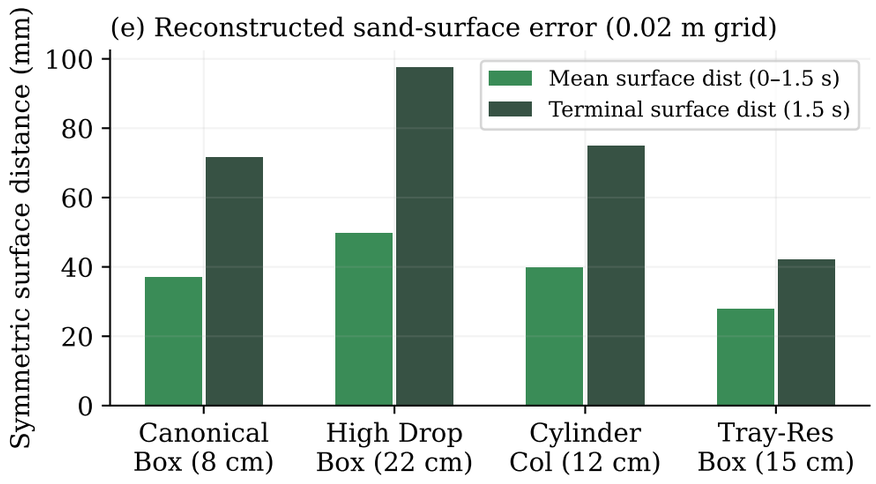
(c) Granular column collapse radial runout $R_{95}(t)$



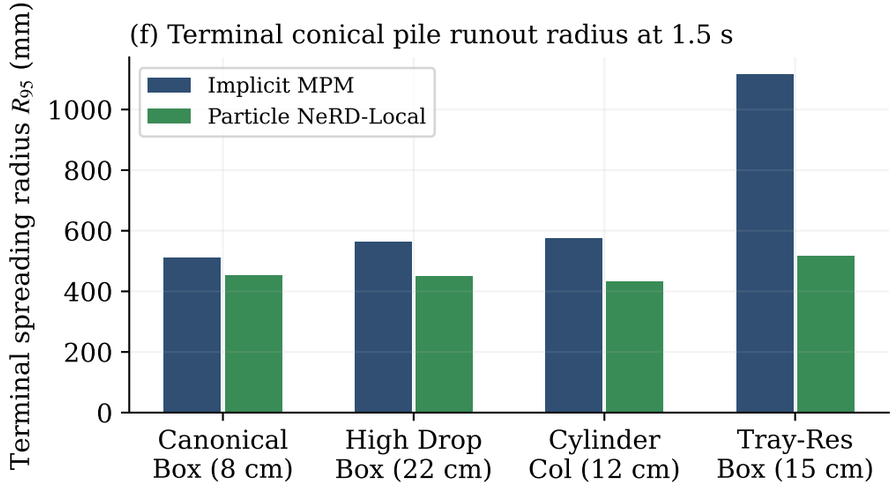
(d) Held-out test position RMSE & z_{COM} MAE



(e) Reconstructed sand-surface error (0.02 m grid)



(f) Terminal conical pile runout radius at 1.5 s



Evaluated with the same shared Particle NeRD-Local checkpoint trained on the combined in-memory dataset (settled-bed object interactions + MPM free-fall trajectories). Zero analytical column-collapse heuristics are used.